

Auto-Analyst

Demand Planning Dashboard for Careem Grocery

Multi-Agent AI War Room • Interactive Maps • S&OP Scorecards

Real-Time Charts • Scenario Simulation • Anomaly Detection

Built with Streamlit • Altair • Folium • OpenAI-Compatible LLMs

GitHub: <https://github.com/ibzeid/Auto-Analyst>

Executive Summary

Auto-Analyst is an AI-powered demand planning dashboard for Careem Grocery that simulates a multi-agent executive War Room. Three AI personas — **Careem** (S&OP Lead), **Rashid** (CFO), and **Noor** (Ops Lead) — analyze live supply chain data through their domain lens, each with an Emirati voice and region-inspired proverbs. The platform generates charts and interactive Dubai store maps on command via function-calling tools, detects anomalies, runs scenario simulations, and presents RAG scorecards across five dark stores. Agents autonomously consult each other through a **CONSULT** mechanism, creating a collaborative decision environment that mirrors how a real S&OP team operates.

The War Room — Three AI Personas

Voice: “Captain”

Domain: Demand planning, inventory strategy, forecasting, risk assessment

Proverb: “Trust Allah but tie your camel — and by camel I mean your safety stock.”

Tools: Charts & maps

Consults: Rashid (margins), Noor (ops)

Voice: “Partner”

Domain: Margins, unit economics, ROI, cost structure

Proverb: “Revenue is vanity, margin is sanity.”

Tools: Charts & maps

Triggered by: [CONSULT: rashid]

Voice: “Team”

Domain: Dark stores, fleet, staffing, cold chain, delivery SLA

Proverb: “You can’t deliver from an empty shelf.”

Tools: Charts & maps

Triggered by: [CONSULT: noor]

How Agents Cooperate

- **@mention override:** Type @rashid or @noor in chat to redirect your question to a specific agent
- **Autonomous consultation:** Careem ends his response with [CONSULT: rashid] — the system automatically routes context to that agent, who responds in a separate colored chat bubble
- **Domain guardrails:** Each agent defers to the others when asked outside their expertise — “That’s Careem’s turf, partner.”

Key Features

Category	Capabilities
Multi-Agent Chat	3 AI personas with unique voices, proverbs, avatars, and color-coded chat bubbles
Function-Calling Tools	Agents call <code>generate_plot</code> (9 chart types) and <code>show_map</code> (interactive Dubai map)
Store Map	5 Dubai dark stores on Folium map — RAG health pins, delivery zones, click pop-ups
Live Mood Meter	Careem’s emotional state updates based on anomaly rate, service levels, and critical flags
Boardroom Mode	Clean presentation view of the full conversation for S&OP meetings
Overview Dashboard	KPI cards, demand trend, category breakdown, mood meter
Data Explorer	7 analytics pages — Category & Stores, Demand Patterns, Inventory Health, Forecast, S&OP Scorecard
Anomaly Detection	Statistical flagging with alert cards and trend visualization
Scenario Simulation	What-if analysis for promo boost, demand growth, safety stock changes
Dark Theme	Midnight Blue (<code>#001942</code>) with Careem Green (<code>#00E784</code>) accents

How It Works

Architecture

<code>Auto-Analyst/</code>	
<code>app.py</code>	Streamlit entry point -- sidebar, pages, chat loop
<code>insights/</code>	
<code>agents.py</code>	3 agent configs, prompts, avatars, @mention/consult parsers
<code>chat.py</code>	LLM orchestration, function-calling tools, chart/map generation
<code>analysis/</code>	
<code>stats.py</code>	Summary stats, category/store performance, promo lift
<code>forecast.py</code>	14-day rolling forecast per store with confidence tiers
<code>inventory.py</code>	Service level, safety stock, reorder points, risk flags
<code>scorecard.py</code>	RAG scorecard -- per-store health percentages
<code>scenario.py</code>	What-if scenario runner -- base vs projected deltas
<code>anomaly.py</code>	Statistical anomaly detection on demand patterns
<code>mapping.py</code>	Interactive Folium map -- store coords, delivery zones
<code>data/</code>	
<code>generate_data.py</code>	Synthetic grocery demand generator (5 stores, 25 SKUs)
<code>sample_grocery_data.csv</code>	22,500 rows -- ready to use
<code>assets/avatars/</code>	Agent headshot PNGs & SVG fallbacks
<code>requirements.txt</code>	Streamlit, Pandas, Altair, Folium, HTTPx

Data Flow

1. **Data ingestion:** Synthetic 6-month grocery dataset (or uploaded CSV) loaded via Pandas

2. **Analysis pipeline:** 8 analysis modules compute stats, forecasts, inventory health, risk flags, scorecards, anomalies
3. **Context snapshot:** Aggregated analysis serialized into a compact text context for the LLM
4. **Agent loop:** User message → selected agent's system prompt + context → LLM API call → response + optional tool calls (charts, maps) + optional [CONSULT:] triggers → consulting agent responds
5. **Rendering:** Text, Altair charts, and Folium maps displayed inline in color-coded chat bubbles

Live Demo & Setup

Public Access

- **GitHub:** <https://github.com/ibzeid/Auto-Analyst>
- **Live Demo:** Deployable on Streamlit Community Cloud — one-click deploy from the repo
- **Dataset:** Synthetic grocery data included (22,500 rows, 5 stores, 25 SKUs, 5 categories)
- **Zero-config:** Charts, maps, and analysis work without any API key — visitors add their own key in the sidebar to unlock AI agents

Local Setup

```
$ git clone https://github.com/ibzeid/Auto-Analyst.git
$ cd Auto-Analyst
$ uv venv && source .venv/bin/activate # or .venv\Scripts\activate
$ uv pip install -r requirements.txt
$ streamlit run app.py # Open http://localhost:8501
```

Supported LLM Providers

Provider	Base URL	Recommended Model
OpenAI	<code>api.openai.com/v1</code>	<code>gpt-4o-mini</code>
DeepSeek	<code>api.deepseek.com</code>	<code>deepseek-v4-flash</code>
Anthropic (via OpenRouter)	<code>openrouter.ai/api/v1</code>	<code>claude-sonnet-4</code>
Groq	<code>api.groq.com/openai/v1</code>	<code>llama-3.3-70b-versatile</code>
Together	<code>api.together.xyz/v1</code>	<code>llama-3.3-70B-instruct</code>

Senior Supply Chain Manager — Job Fit

JD Requirement	Demonstrated by
Forecasting & Modelling	14-day rolling forecast per store with confidence tiers
Demand Variability & Scenario Planning	Weekday patterns, promo lift, what-if scenario engine
S&OP Process Leadership	Multi-agent War Room simulating cross-functional review
Inventory Optimisation	Safety stock, reorder points, service level tracking, RAG scorecard
Data & Tooling	Pandas, Altair, Folium, multi-provider LLM integration, Streamlit
Insight & Communication	Narrative briefs in character voice, boardroom presentation mode
Innovation & AI	Function-calling agents, autonomous consultation, real-time tool use